Pathogenesis, Clinical Manifestations, and Natural History of Diabetic Kidney Disease

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INTRODUCTION

Diabetic kidney disease (DKD) is the leading cause of kidney failure in high-income countries. It can develop in the course of either type 1 or type 2 diabetes mellitus (DM). *Type 1 diabetes* (T1D) is an autoimmune disease characterized by antibody-mediated and cell-mediated destruction of pancreatic islets. T1D may occur at any age but usually presents before the age of 30 years. *Type 2 diabetes* (T2D) is characterized by a combination of insulin resistance and insulin deficiency. The *metabolic syndrome* (insulin resistance, visceral obesity, hypertension, hyperuricemia, and dyslipidemia) is often followed by T2D. For a long period, insulin resistance is compensated by increased insulin secretion, but a gradual decline in pancreatic β-cell mass and function finally culminates in hyperglycemia, and patients with T2D may require treatment with insulin. T2D is increasingly seen in younger adults, adolescents, and even children. Other types of DM include maturity-onset diabetes of the young, gestational diabetes, diabetes secondary to various metabolic disorders, and iatrogenic diabetes resulting from corticosteroid or other immunomodulatory treatments.

PATHOGENESIS OF DIABETIC KIDNEY DISEASE

Genetic and Environmental Factors

The risk of developing DKD is equal in T1D and T2D, and only 30% to 40% of patients with T1D or T2D will ultimately develop nephropathy. The prevalence of nephropathy in diabetic patients varies among different racial and ethnic groups such that it is relatively increased in African Americans, Native Americans, Mexican Americans, Polynesians, Aboriginal Australians, and urbanized South Asian immigrants in the United Kingdom compared with Whites. Although barriers to care likely account for some of these interpopulation differences, polygenetic factors likely also contribute.

Familial clustering of DKD has been reported in T1D and T2D and in White and non-White populations. In a person with T1D who has a first-degree relative with diabetes and nephropathy, the risk for development of DKD is 83%. The frequency is only 17% if there is a first-degree relative with diabetes but without nephropathy.[1] In T2D, familial clustering has been well documented in Pima Indians.[2] A familial determinant is also suggested by higher albumin excretion rates in offspring of patients with T2D with nephropathy. The risk is particularly high in the offspring if the mother had been hyperglycemic during pregnancy, perhaps because this causes reduced formation of nephrons ("nephron underdosing") in the offspring.[3,4] Low birthweight and nephron underdosing are also associated with hypertension, metabolic syndrome, and perhaps DKD. Nephron underdosing[5] is believed to lead to compensatory glomerular hypertrophy and increased single-nephron glomerular filtration rate (GFR), thus aggravating glomerular injury in diabetes.

The risk for DKD does not follow simple mendelian inheritance, and multiple genes are presumably involved. In patients with T1D the estimate of heritability for nephropathy was 35%; however, replication studies did not identify any single genetic variances reaching whole-genome levels of significance.[6] Nevertheless, single nucleotide polymorphisms (SNPs) in the engulfment and cell motility 1 *(ELMO1)* gene have been associated with risk for DKD in several ethnic groups with T2D.[7-9] Gene polymorphisms with 300,000 SNPs were analyzed in a large T1D cohort (over 6000 individuals with T1D and their relatives) and recently identified the protein coding gene *ARHGAP24*, associated with focal glomerulosclerosis,[10] as linked to DKD. Gene polymorphisms also may contribute to familial clustering.

Environmental factors, especially diet, may be involved in the pathogenesis of diabetes and DKD. One of the strongest risk factors is the intake of soft drinks containing added sugars such as sucrose or high-fructose corn syrup. Fructose increases uric acid levels, a predictor for the development of T2D and potentially also DKD,[11] probably via uric acid–induced oxidative stress and endothelial dysfunction. However, a randomized study with allopurinol in patients with T1D and early to moderate DKD failed to reduce the rate of progression despite lowering uric acid levels.[12] Smoking is a strong risk factor for progression of DKD and may be related to hypoxia in the kidney. Other putative associations include sleep apnea,[13] higher caloric intake, and lower levels of exercise.

Hemodynamic Changes

Hyperfiltration is common in early diabetes but can be corrected with good glycemic control. Increased GFR involves glucose-dependent effects causing afferent arteriolar dilation mediated by a range of vasoactive mediators, including insulin-like growth factor 1 (IGF-1), transforming growth factor-β (TGF-β1), vascular endothelial growth factor (VEGF), nitric oxide (NO), prostaglandins, and glucagon (Fig. 31.1). Over time, vascular disease of the afferent arteriole may result in permanent alterations in kidney autoregulation that favor glomerular hypertension. Kidney injury in DKD is caused not only by hemodynamic disturbances (e.g., hyperfiltration, hyperperfusion) but also by disturbed glucose homeostasis, and the two pathways interact. For example, shear stress increases glucose transport into mesangial cells by upregulation of specific glucose transporters. Furthermore, shear stress and mechanical strain resulting from altered glomerular hemodynamics trigger autocrine and paracrine release of cytokines and growth factors in the glomerulus.

DKD is also associated with tubular abnormalities in which angiotensin II (Ang II) causes hypertrophic proximal tubular growth and increased sodium reabsorption.[14] Specific inhibition of the sodium glucose cotransporter-2 (SGLT2) in proximal tubular cells slowed progression of DKD, highlighting the role of tubuloglomerular feedback and glomerular hyperfiltration in DKD (see Chapter 32).[15] shiftenterInsulin itself stimulated matrix production in proximal tubular epithelial cells (TECs), and deficiency of the insulin receptor specifically in proximal TECs protected against obesity-induced kidney disease,[16] implicating a key role of elevated insulin levels in kidney disease progression.

Glomerular Changes

Kidney growth occurs early after the onset of diabetes. Glomerular enlargement is associated with increased mesangial cell number, hypertrophy, and increase of capillary loops, thus enhancing the filtration surface area. Renal tubular hypertrophy is primarily the result of TEC proliferation and hypertrophy.

Experimentally, avoidance of hyperglycemia prevents kidney hypertrophy. Hyperglycemia causes hypertrophy by stimulating growth factors in the kidney, including IGF-1, epidermal growth factor (EGF), platelet-derived growth factor (PDGF), VEGF, TGF-β, and Ang II. Hyperglycemia also induces thrombospondin, a potent activator of latent TGF-β. Antagonizing TGF-β in diabetic patients, however, did not confer beneficial effects on kidney function or proteinuria.[17] Similarly, inhibition of VEGF prevented glomerular hypertrophy in models of DKD and reduced albuminuria.[18] Urine EGF levels are linked to progression of chronic kidney disease (CKD) of multiple etiologies, including DKD.[19]

The pathologic hallmarks of diabetic nephropathy (DN) are mesangial expansion, nodular diabetic glomerulosclerosis (the acellular Kimmelstiel-Wilson lesion), and diffuse glomerulosclerosis. Mesangiolysis likely plays a key role in cell loss and nodule formation. Increasing evidence suggests that local NO deficiency contributes to these histologic lesions, in particular nodule formation.

The mechanisms underlying the emergence of albuminuria start with widening of the glomerular basement membrane (GBM) with accumulation of type IV collagen and net reduction in negatively charged heparin sulfate proteoglycan (see the section Kidney

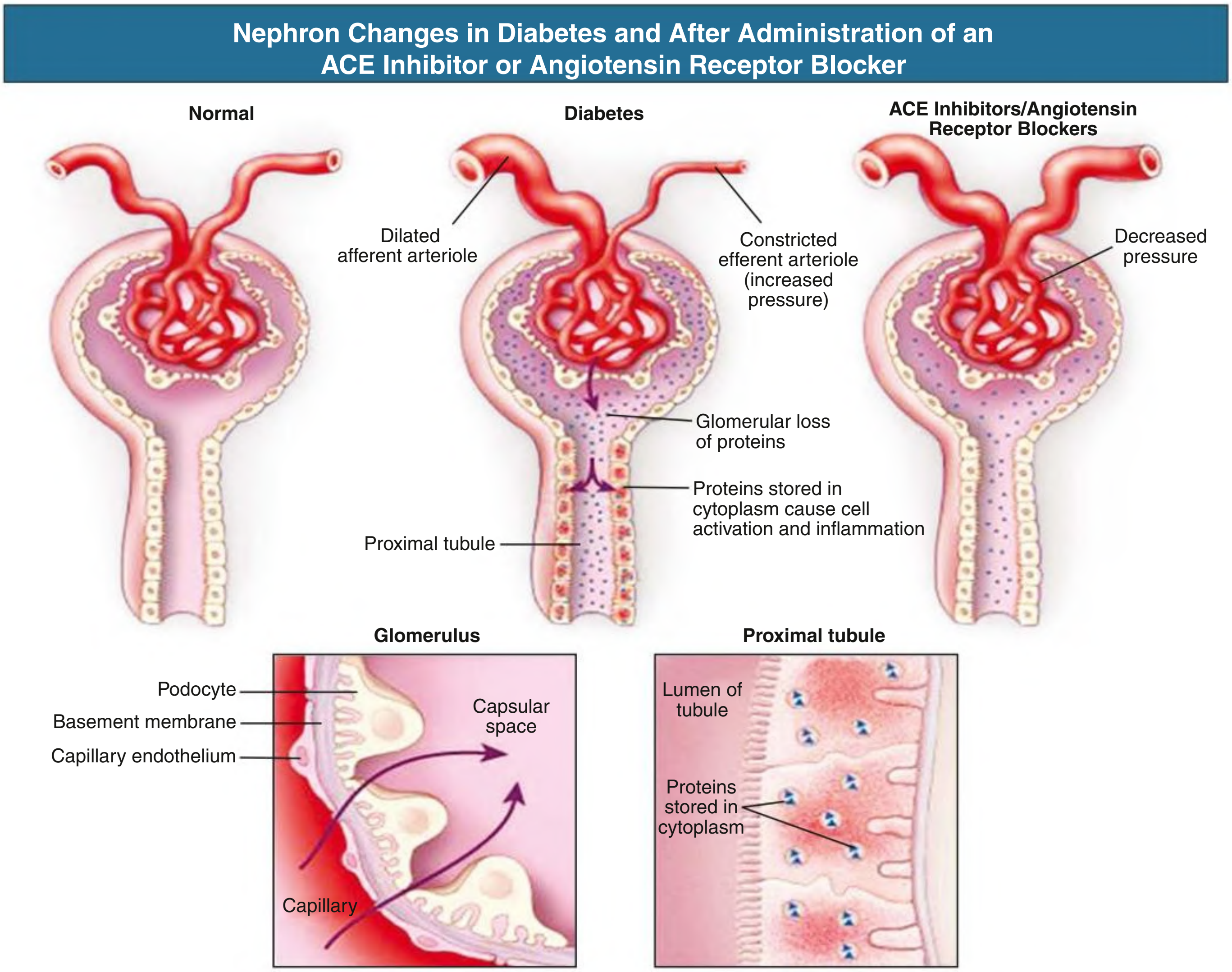

Fig. 31.1 Comparison of normal nephron, nephron in diabetic kidney disease (DKD), and nephron in DKD after angiotensin-converting enzyme (ACE) inhibitor/angiotensin receptor blocker (ARB) administration. Note afferent vasodilation and efferent angiotensin II–mediated vasoconstriction in DKD causing glomerular hypertension, which is relieved by an ACE inhibitor and ARB therapy. Note also protein leakage into the filtrate and tubular loading, with endocytosed protein causing an inflammatory reaction that promotes interstitial fibrosis. This is reversed by ACE inhibitor/ARB treatment.

Pathology). The expression of one permeability-controlling protein, nephrin, is abnormally low in DKD.[18] Its transcription is suppressed by Ang II and restored by renin-angiotensin system (RAS) blockers. Apoptosis of podocytes is triggered by various factors, including Ang II and TGF-β, and adhesion of podocytes to the GBM is reduced by advanced glycation end products (AGEs)-induced suppression of neuropilin-1. Podocyte loss also follows hyperglycemia-induced reactive oxygen species (ROS) generation, causing podocyte apoptosis or detachment. ROS generation in podocytes may be largely mediated by Nox4,[20,21] although other contributors of ROS have also been implicated. Migration of podocytes is also attenuated by the reduction of neuropilin-1, thereby preventing surviving podocytes from covering denuded areas of GBM, which promotes development of focal segmental glomerulosclerosis (FSGS).

Crosstalk between glomerular endothelial cells and podocytes involves activated protein C (APC), the formation of which is regulated by endothelial thrombomodulin and is reduced in diabetic mice.[22] In DKD, thrombomodulin-dependent APC formation inhibits podocyte apoptosis.[23] In cultured podocytes, administration of the adipocyte-specific hormone adiponectin prevented high glucose–induced podocyte dysfunction. Endothelial cell dysfunction associated with altered fenestrations and glycocalyx may contribute to enhanced permeability. Adiponectin levels are low in patients with the metabolic syndrome or T2D, which may contribute to the development of albuminuria[20]; however, once diabetes is established, elevated adiponectin levels correlate with faster progression of disease. In advanced DKD, albuminuria evolves into nonselective proteinuria with high-molecular-weight serum proteins escaping the GBM with disrupted texture, gaps, and holes.

Tubular and Inflammatory Changes

Tubulointerstitial injury ultimately determines the rate of attrition of kidney function. In vitro studies demonstrate the pathogenic role of various diabetic substrates in promoting tubule hypertrophy, extracellular matrix (ECM) production, and inflammation,[18] which is a complex process with a large number of interacting pathways that lead to a chronic inflammatory infiltrate, which includes macrophages and other immune cells that release cytokines and profibrotic factors and interact with intrinsic kidney cells to create a profibrotic microenvironment. Chemokines and their receptors, in particular monocyte chemotactic protein-1 (MCP-1/CCL2), RANTES/CCL5, IL-6, and tumor necrosis factor (TNF) receptors, as well as adhesion molecules (e.g., ICAM-1), contribute to persistent inflammation that triggers a profibrotic cascade in the kidney (Fig. 31.2).[18,24] Glomerular cells, TECs, macrophages/lymphocytes, and fibroblasts/myofibroblast all contribute to matrix accumulation along the glomerular and tubular basement membranes and within the interstitium. Myofibroblasts promote progression of fibrosis in DKD by facilitating deposition of interstitial ECM (Fig. 31.3). Soluble TNF receptors appear to be a robust biomarker for progressive kidney disease in both T1D and T2D. Th17 immune cells and their effector cytokine IL-17A take part in diabetes-mediated kidney damage and could be a promising therapeutic target.[25]

Bardoxolone methyl, as an inducer of the KEAP1-Nrf2 pathway, exhibits antiinflammatory effects. Although the BEACON trial was terminated due to safety concerns about early heart failure, a post hoc analysis[26] revealed reduction in albuminuria coupled to estimated glomerular filtration rate (eGFR) reduction. The Japanese TSUBAKI study trial showed that bardoxolone methyl increased GFR among 65 patients with T2D and stage 3 or 4 CKD versus placebo.[27] The oral CCR2 inhibitor CCX140-B for 52 weeks reduced residual albuminuria in subjects with T2D.[28] The FIDELIO-DKD study[29] found that finerenone, a nonsteroidal blocker of the mineralocorticoid receptor (half-life ~2 hours and without metabolites), reduced the rate of progression of DKD in T2D without causing hyperkalemia. Any beneficial effects of finerenone may be mediated by reductions in inflammation, as its blood pressure lowering effects were minimal.

Clinical studies showed that even mild anemia (hemoglobin level <12.5 g/dL for men, <11.5 g/dL for women) increases the risk for progression of DKD. Anemia presumably causes kidney hypoxia. Moreover, hypoxia is exacerbated by the progressive hyalinosis of the afferent and efferent arterioles and loss of peritubular capillaries. In experimental chronic kidney injury, hypoxia is an important factor aggravating interstitial fibrosis. The transition of TECs into fibroblasts is stimulated by cellular hypoxia.[30] The induction of growth factors and cytokines is mediated by hypoxia-inducible factor-1 (HIF-1), which can be amplified by Ang II. However, the TREAT trial[31] showed that anemia treatment with darbepoetin-alfa initiated at a hemoglobin level of around 10.5 g/dL does not slow the rate of GFR loss in both diabetic and nondiabetic CKD.

Hyperglycemia

Role of Glucose Control

The following clinical evidence sheds light on the role of tight glycemic control in slowing the development of DKD:

- In the Diabetes Control and Complications Trial (DCCT), there was a remarkable reduction in progression from normoalbuminuria to moderately increased albuminuria and other microvascular complications, specifically retinopathy, in patients with T1D with tight glycemic control.[32]
- Euglycemia that followed isolated pancreatic transplantation was associated with regression of diabetic glomerulosclerosis after 10 years.[33]
- In the United Kingdom Prospective Diabetes Study (UKPDS), reducing the hemoglobin A_{1c} (HbA_{1c}) level by approximately 0.9% in patients with T2D reduced the risk for microvascular complications, including nephropathy.[34]
- In the ADVANCE study, intensive glucose control with a target HbA_{1c} level of 6.5% was associated with a long-term reduction in kidney failure, without evidence of any increased risk for cardiovascular events or death.[35]
- In the EMPA-REG study, individuals with type 2 diabetes treated with empagliflozin with lower glycated hemoglobin levels had significant reduction in cardiovascular and kidney events, though the protective effects of the SGLT2 inhibitor are likely beyond simply glycemic control.[15]
- In the CREDENCE trial,[36] canagliflozin reduced the risk of kidney failure and cardiovascular events among 4401 individuals with type 2 diabetes with an eGFR of 30 to 90 mL/min per 1.73 m^2 and albuminuria 300 to 5000 mg/g after a median follow up of 2.62 years. These benefits were consistent across eGFR subgroups, including those initiating treatment at eGFR 30 to 45 mL/min per 1.73 m^2.[37]
- In the DAPA-CKD trial,[38] dapagliflozin conferred kidney and cardiovascular protection among patients with diabetic and nondiabetic CKD with eGFR 25 to 75 mL/min per 1.73 m^2 and albuminuria 200 to 5000 mg/g, suggesting that the clinical benefits of SGLT2i are a class effect that extends to people with CKD. The larger EMPA-KIDNEY trial (NCT03594110), which recruited patients with eGFR down to 20 mL/min/1.73 m^2 will further address the latter possibility.

A unifying theory was that hyperglycemia leads to diabetic complications through an accumulation of mitochondrial superoxide production and oxidative sequelae.[39] As the kidneys are highly active metabolically and require vast amounts of adenosine triphosphate (ATP) for their normal function, they are

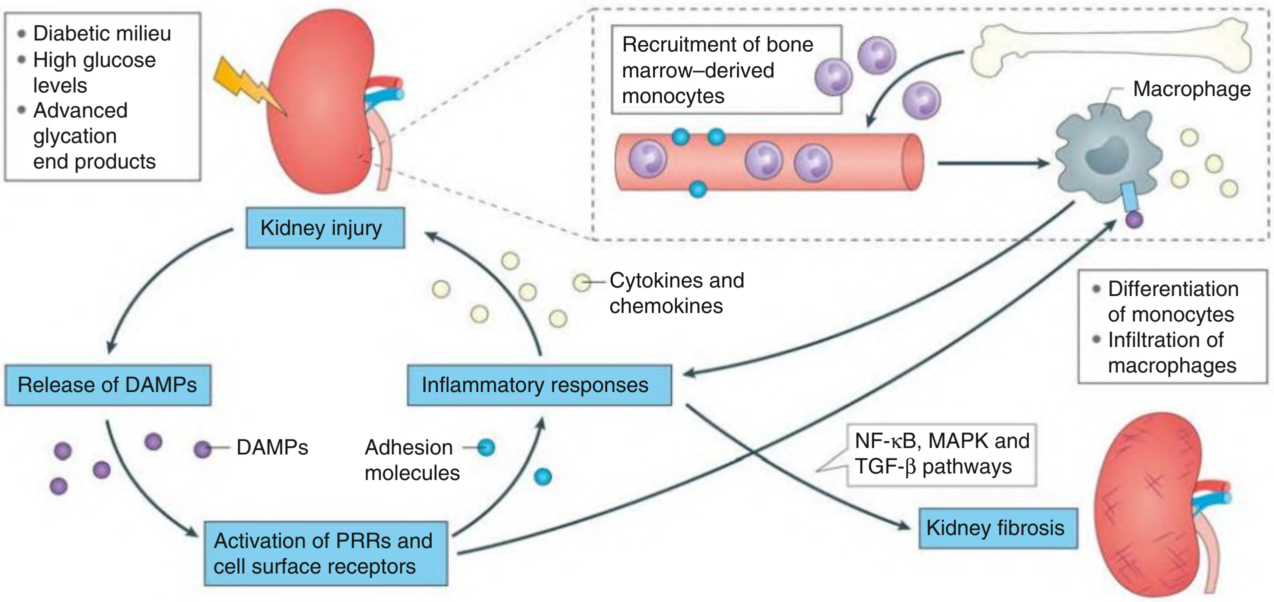

Fig. 31.2 **Crosstalk Between Endothelial Cells and Podocytes Involving Protein C.** Under physiologic conditions, protein C is activated by the binding of thrombin to its cofactor, thrombomodulin, on glomerular endothelial cells. The formed complex catalyzes the conversion of protein C to its catalytically activated form, which has potent anticoagulant, profibrinolytic, antiinflammatory, and cytoprotective effects. In DKD, the production of activated protein C (APC) in the glomerulus is reduced because of suppression of thrombomodulin expression. Decreased functional activity of APC affects the permeability of the glomerular capillary wall and enhances apoptosis of glomerular endothelial cells and podocytes. (From Gilbert RE, Marsden PA. Activated protein C and diabetic nephropathy. *N Engl J Med.* 2008;358:1628–1630.)

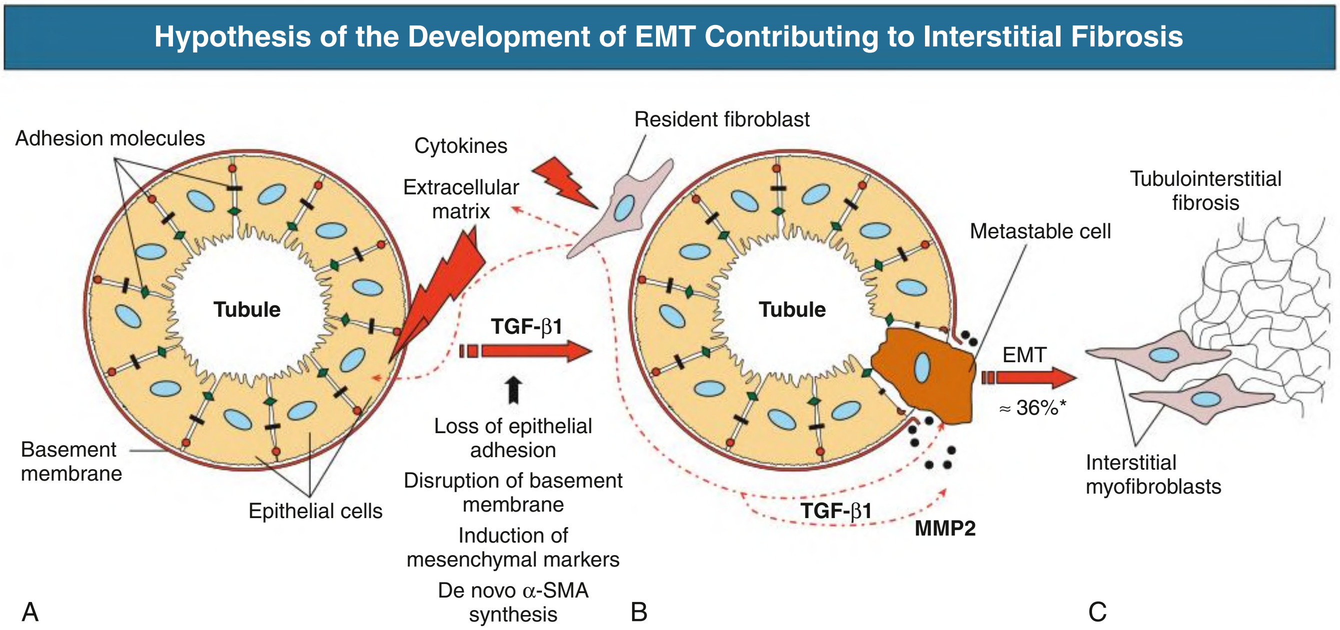

Fig. 31.3 **Hypothesis of the Development of Epithelial-Mesenchymal Transition (EMT) Contributing to Interstitial Fibrosis.** Initiated by external stimuli (e.g., cytokines), tubular cells lose their cell-cell contacts (e.g., E-cadherin) (A) and start to express mesenchymal markers (e.g., α-SMA, vimentin) (B). After disruption of tubular basement membrane (by MMP2), metastable cells disengage themselves from cell connective and transdifferentiate to interstitial myofibroblasts that synthesize extracellular matrix and contribute to fibrosis (C). *MMP,* Matrix metalloproteinase; *SMA,* smooth muscle actin; *TGF-β1,* transforming growth factor-β1. *Up to 36% of all interstitial myofibroblasts in diabetic kidney disease are thought to derive from EMT. (Modified from Loffler I. Pathophysiology of diabetic nephropathy. In: Wolf G, ed. *Diabetes and Kidney Disease.* Wiley-Blackwell; 2013.)

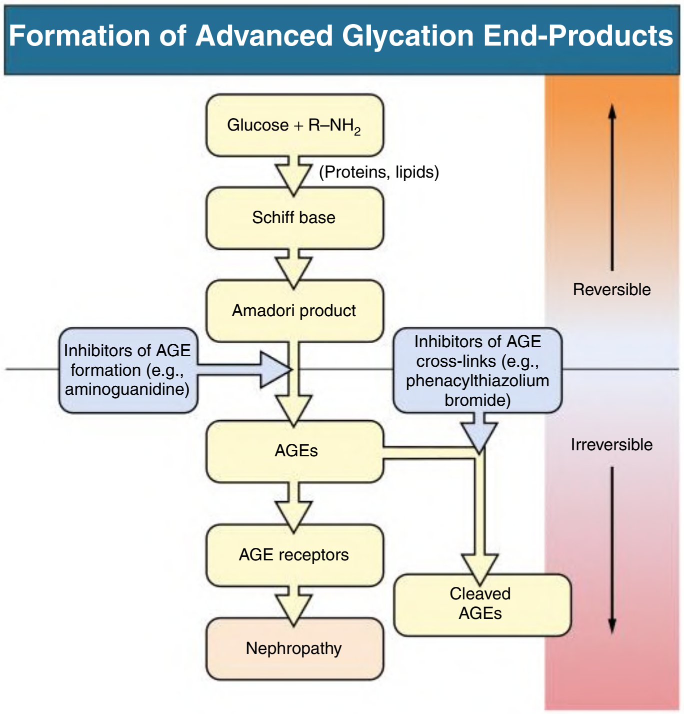

Fig. 31.4 Mechanism of formation of advanced glycation end-products (AGEs).

mitochondrially rich. Diabetes alters the delivery of substrates and oxygen to the kidneys, and changes in metabolic fuel sources to meet ATP demands result in increased oxygen consumption, which contributes to kidney hypoxia. Mitochondrial fission and fusion events enable energy demands to be met and provide mitochondrial quality control. Disruption of these events in diabetes reduces mitochondrial biogenesis, as reflected by a deficiency of the master regulator of mitochondrial biogenesis peroxisome proliferator-activated receptor γ coactivator 1α.[40] Restoring mitochondrial biogenesis and mitochondrial superoxide levels is associated with reduced inflammation, albuminuria, and fibrosis in experimental DKD.[41] The protective effect of SGLT2 inhibitors may be mediated by normalizing mitochondrial function,[42] and enhancing SIRT1 and hypoxia inducible factor 2α (HIF-2α) signaling[43] that can explain the biologic consequences of ketonemia and erythrocytosis. Recent studies using urine metabolomics have identified that mitochondrial dysfunction plays a role in progression of DKD,[44-46] and one recent clinical study[47] demonstrated that SGLT2 inhibitors have beneficial effects on markers of mitochondrial function. There is likely an important role for hepatic ketone generation in response to SGLT2 inhibition, which may be a preferential fuel for kidney proximal TECs and cardiac myocytes.

Pathways Related to Glucose Metabolism

Protein kinase C (PKC) is a family of serine-threonine kinases that regulate diverse vascular functions. PKC-α mediates diabetic podocyte injury via phosphorylation of p66SHC and its inhibition by Klotho overexpression attenuated podocyte injury and proteinuria in streptozocin-treated mice.[48] Protein kinase C–η mediates advanced oxidation protein product–induced mitochondrial dysfunction and oxidative stress in kidney tubular cells.[49]

Chronic hyperglycemia leads to nonenzymatic glycation of amino acids and proteins (Maillard or Browning reaction)[50] (Fig. 31.4).

AGEs are increased in the serum, glomeruli, and tubules of DKD patients. They bind to macrophages, mesangial cells, and tubular cells and mediate cellular actions, including expression of adhesion molecules, cell hypertrophy, ECM synthesis, epithelial-mesenchymal transition (EMT), and inhibition of NO synthetase (NOS). AGEs injected in vivo induce albuminuria and glomerulosclerosis.[50] AGEs induce podocyte hypertrophy, followed by apoptosis and suppression of nephrin synthesis. Among several binding sites, the most important is RAGE (receptor for advanced glycation end product), which is present in tubular cells and podocytes. Ang II stimulates upregulation of RAGE on podocytes.[51] This effect is mediated by Ang II (AT_2) receptors not blocked by sartanes.[51] One of the actions of RAGE is activation of nuclear factor-κB (NF-κB). The soluble extracellular domain of RAGE (sRAGE) acts as a decoy receptor and experimentally ameliorates kidney lesions in diabetes.[50] In T1D, sRAGE is associated with progression from severely increased albuminuria to kidney failure, though causality could not be established.[52] In T2D, sRAGE is an early predictor of renovascular complications.[53]

Blocking the binding of AGEs to its receptor using a RAGE-aptamer could inhibit tubular injury in diabetic mice partly by suppressing the AGE-RAGE-oxidative stress axis and improving insulin resistance independent of glycemic control.[54] A selective RAGE inhibitor, FPS-ZM1, combined with valsartan alleviated podocyte injury in streptozocin-treated rats by restoring nephrin and synaptopodin expression and lowering albuminuria and serum cystatin C.[55]

The AGE/RAGE interaction upregulated myo-inositol oxygenase (MIOX), a tubule-specific enzyme modulating redox balance, to induce ROS generation and NF-κB activation to cause tubulointerstitial injury.[56] Dietary supplementation of d-glucarate in diabetic mice decreased MIOX expression, attenuated tubular damage, and improved kidney function by attenuating mitochondrial fragmentation, oxidative stress, and apoptosis and restoring autophagy/mitophagy in tubular cells.[57] In transgenic models, MIOX overexpression accentuated, whereas its knockdown protected the diabetic kidney in mice from tubulointerstitial injury.[58]

Finally, some fructose-6-phosphate is diverted into the hexosamine pathway, increasing the concentrations of *N*-acetylglucosamine, which modifies certain transcription factors, such as Sp1 activity. In turn, Sp1 upregulates key fibrotic mediators, such as TGF-β1 and plasminogen activator inhibitor 1.

Adenosine Monophosphate Kinase

The energy-sensing enzyme 5′-adenosine monophosphate kinase (AMPK) shiftenterpathway may contribute to DKD. Inhibition of 5′-AMP-activated protein kinase in caloric-excess states has been linked to inflammation (NF-κB, nicotinamide adenine dinucleotide phosphate [NADPH] oxidase, monocyte chemotactic protein-1 [MCP-1] stimulation), vascular dysfunction (endothelial NOS inhibition), stimulation of hypertrophy (mammalian target of rapamycin complex [activation]), and profibrotic pathways (TGF-β signaling).[59-61] Stimulation of AMPK is beneficial in both type 1 and type 2 models of DKD. Key regulators of mitochondrial health-adenosine monophosphate kinase, sirtuins, and PCG1α (peroxisome proliferator-activated receptor γ coactivator-1α) have all been shown to play significant roles in the resilience of the kidney against disease.[62]

Activation of Innate Immunity

Diabetes can be considered a metabolic danger signal that is effected via toll-like receptors (TLRs), a conserved family of pattern recognition receptors that play a fundamental role in innate immunity.[24] In particular, TLR-4 is overexpressed in the kidney tubules of human DN biopsies.[63,64] In vitro, silencing TLR-4 ameliorated high glucose–induced tubular cell inflammation. In experimental DKD, either systemic deletion or the application of a TLR-4 antagonist conferred renoprotection.[64] In addition, TLRs are also expressed in other resident kidney cell types such as podocytes and mesangial cells, as well as infiltrating macrophages, that could act in concert to bring about an inflammatory phenotype observed in DKD (Fig. 31.5). The NLRP3 inflammasome links sensing of metabolic stress in the diabetic kidney to activation of proinflammatory cascades via the induction of IL-1β and IL-18. The kallikrein-kinin system promotes inflammatory processes via the generation of bradykinins and the activation of bradykinin receptors, and activation of protease-activated receptors on kidney cells by coagulation enzymes contributes to kidney inflammation and fibrosis in DKD.

A key component of innate immunity includes the NADPH oxidase pathway, expressed primarily in the lysosomes of phagocytic cells, but also in the kidney. Enhanced Nox4 in podocytes may contribute to glomerular disease involving a pathway linking Nox to the citric acid cycle.[65]

Complement has been increasingly implicated in DKD. Increased levels of C5a were detected in kidney tubules from patients with biopsy confirmed DN and its intensity correlated with the progression of the disease.[66] In experimental DKD, genetic deletion of C5aR1 in mice conferred protection against diabetes-induced kidney injury and in vitro, C5a/C5aR propagated injury during DKD by disrupting mitochondrial agility.[67]

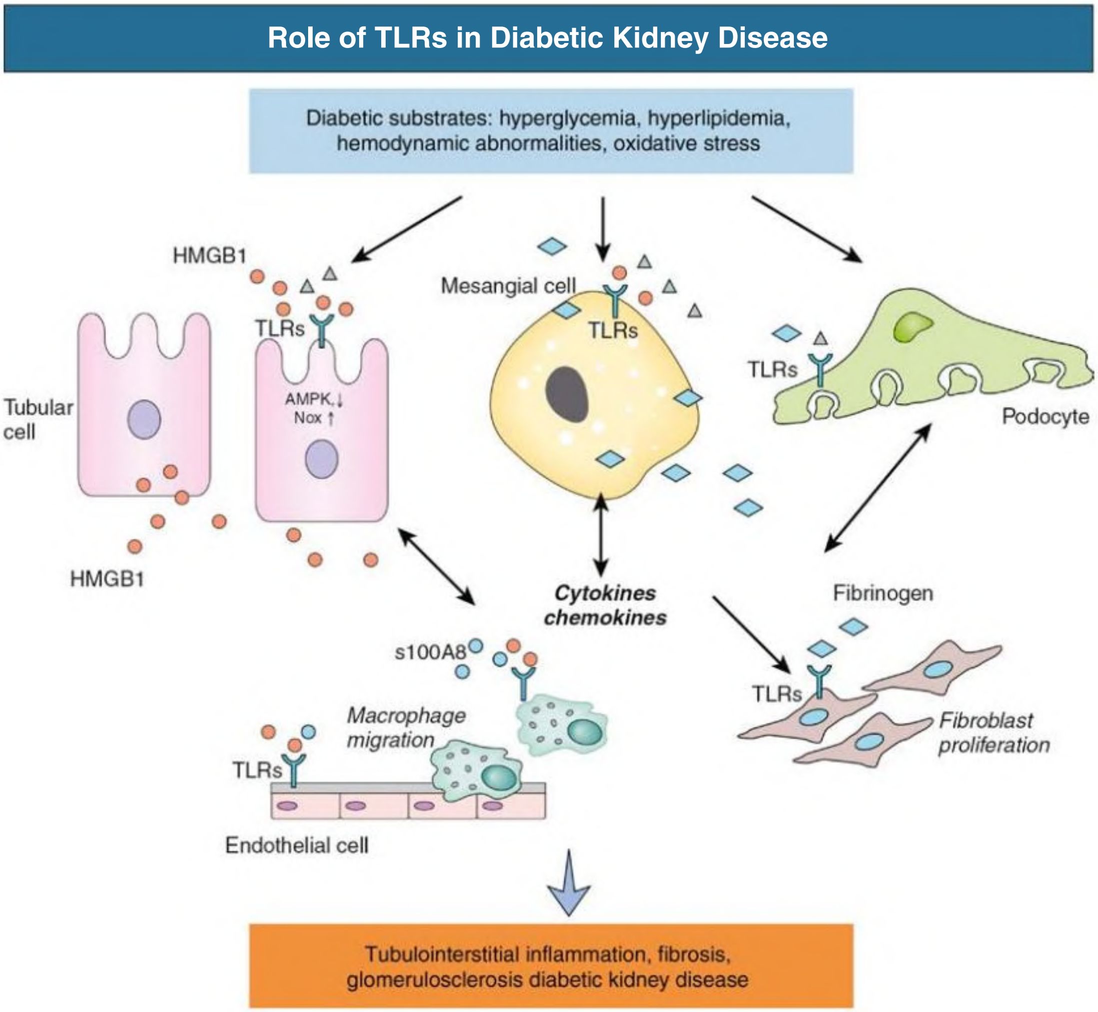

Fig. 31.5 Role of Toll-like Receptors (TLRs) in Diabetic Kidney Disease. Toll-like receptors in resident kidney cells could recognize and respond to the metabolic stress of diabetes or endogenous ligands activated during the diabetic state, inducing downstream signaling events to propagate the synthesis of proinflammatory cytokines and chemokines, which act as effectors to further facilitate macrophage recruitment and fibroblast proliferation, leading to a self-perpetuating cycle of kidney inflammation and subsequent tubulointerstitial fibrosis and glomerulosclerosis. *AMPK,* 5′-Adenosine monophosphate kinase; *HMGB1,* high-mobility group box-1 protein; *S100A8,* S100 calcium binding protein A8. (Modified from Lin M, Yiu WH, Wu HJ, et al. Toll-like receptor 4 promotes tubular inflammation in diabetic nephropathy. *J Am Soc Nephrol.* 2012;23:86–102.)

Lipotoxicity

Lipotoxicity is increasingly implicated in the pathogenesis of DKD. In obesity, a myriad of metabolic disturbances including chronic systemic low-grade inflammation and insulin resistance are directly or indirectly associated with not only obesity, but also other metabolic diseases like metabolic syndrome, obesity-related T2D, nonalcoholic fatty liver disease, and cardiovascular disease. Animal and in vitro studies have identified saturated fatty acids as the dominant nonesterified fatty acids in the circulation of obese subjects that act as nonmicrobial agonists to trigger the inflammatory response via activating TLR4 signaling.[68] Increased lipogenesis predicted DKD in T2D.[69] In experimental DKD, kidney VEGF-B expression correlates with the severity of disease and inhibiting its signaling reduces kidney lipotoxicity, resensitizes podocytes to insulin signaling, inhibits the development of diabetic pathologies, and prevents kidney dysfunction.[70]

Epigenetics and Epigenomics

Epigenetic mechanisms involve chromatin histone modifications, DNA methylation, and noncoding RNAs.[71] The first miRNA shown to have a functional role in DKD was miR-192, which acts by targeting key repressors to promote the expression of ECM and collagen and to augment the profibrotic effects of TGF-β1. Numerous miRNAs are now thought to regulate key features of DKD, such as podocyte apoptosis, ECM accumulation, glomerular and tubular hypertrophy, and fibrosis.

LncRNAs are long transcripts (>200 nucleotides and up to ~100 kb in length) that have many similarities with mRNAs but lack protein-coding (translation) potential. They have distinct cellular roles that affect various biologic mechanisms and processes. These mechanisms can respond to changes in the environment and mediate persistent expression of DKD-related genes and phenotypes induced by prior exposure to the diabetic milieu despite subsequent glycemic control, a phenomenon called metabolic memory. A number of lncRNAs could play a pivotal role in development and progression of DKD either via direct involvement or as indirect mediators of nephropathic pathways, such as TGF-β1, NF-κB, STAT3, and GSK-3β signaling. Some lncRNAs may therefore become biomarkers for early diagnosis or prognosis of DKD or as therapeutic targets for DKD.

Gut Microbiome

Among patients with T2D with normoalbuminuria or moderately or severely increased albuminuria, the levels of phenyl sulfate, a gut microbiota–derived metabolite, significantly correlated with basal and predicted 2-year progression of albuminuria in patients with moderately increased albuminuria.[72] In experimental diabetes, phenyl sulfate administration induced albuminuria and podocyte damage, and inhibition of tyrosine phenol-lyase, a bacterial enzyme responsible for the synthesis of phenol from dietary tyrosine before it is metabolized into phenyl sulfate in the liver, reduces albuminuria. Applying broad-spectrum oral antibiotics or fecal microbiota transplantation from healthy donors to diabetic rats decreased serum acetate levels and alleviated tubulointerstitial injury,[73] suggesting that gut microbiota reprogramming might become a new strategy for DKD.

Renin-Angiotensin-Aldosterone System and Diabetic Kidney Disease

Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) slow progression of DKD (see Chapter 32). Although plasma renin activity is low in DKD, it is inappropriate in relation to increased extracellular volume and exchangeable sodium, suggesting activation of the RAS.[74] In experimental diabetes, sites of local RAS activation have been identified in glomeruli and TECs.[74] High glucose and AGEs stimulate angiotensinogen and renin expression in various kidney cells, mainly through ROS.[74] Proteinuria further activates the local RAS of tubular cells.

Ang II has many nonhemodynamic effects and mediates cell proliferation, hypertrophy, ECM expansion, and cytokine (TGF-β, VEGF) synthesis.[74] Therefore, ACE inhibitors and ARBs presumably act via hemodynamic and nonhemodynamic actions.

Aldosterone accelerates progression in kidney damage models independently of Ang II. In DKD, aldosterone escape has been linked to progression of proteinuria (see Chapter 32). Aldosterone synthesis is stimulated in DKD, and this steroid hormone stimulates the synthesis of other proinflammatory and profibrogenic cytokines (MCP-1, shiftenterTGF-β).[75] The beneficial effects of the mineralocorticoid receptor blocker finerenone is likely mediated via its antiinflammatory action.

Other vasoactive agents involved in the pathogenesis of DKD include endothelin, NO, the kallikrein-kinin system, and natriuretic peptides. The SONAR trial[76] showed that the selective endothelin-A receptor antagonist atrasentan, administered to 1325 T2D patients with eGFR 25 to 75 mL/min/1.73 m^2 and albuminuria 300 to 5000 mg/g creatinine versus 1323 placebo control subjects reduced the risk of kidney events after a median follow-up of 2.2 years, though at an increased risk of fluid retention and anemia.

Uric Acid and Fructose

An elevated uric acid level can predict the development of DKD. Uric acid is generated during the metabolism of fructose when it induces mitochondrial oxidative stress. In addition to dietary fructose from added sugars, there is increasing evidence that in diabetes, fructose is generated in the kidney, where it is metabolized to uric acid that may mediate kidney injury. Thus, both dietary and endogenous production of fructose may be involved in the development of diabetes and its complications.[77] Lowering of serum urate levels with allopurinol among 267 individuals with T1D with DKD versus 263 placebo control subjects in the PERL[12] (Preventing Early Renal Function Loss) trial, however, did not affect kidney outcomes.

EPIDEMIOLOGY

The US Renal Data System 2020 Annual Report (www.USRDS.org) revealed in 2018 that DKD was the most frequent primary diagnosis accounting for 47.1% of incident patients being started on kidney replacement therapy. The China Kidney Disease Network (CKD-NET) 2016 Annual Data Report[78] published in 2020 revealed DKD to be the top cause of CKD, accounting for 26.7% of CKD. The proportion of diabetes among patients with kidney failure varies considerably among countries, but in many countries DKD is the leading cause of kidney failure.[79] Classic features of DKD were observed in about 60% of diabetic patients (i.e., normal kidney size despite kidney failure; proteinuria >1 g/2 h with or without retinopathy); 13% had an atypical presentation with ischemic nephropathy, and in 27% a known primary kidney disease coexisted with diabetes. An important mode of presentation has become irreversible acute kidney injury (AKI), for example, after administration of nonsteroidal antiinflammatory drugs (NSAIDs), cardiac events, and septicemia. Many patients also lose the clinical manifestations

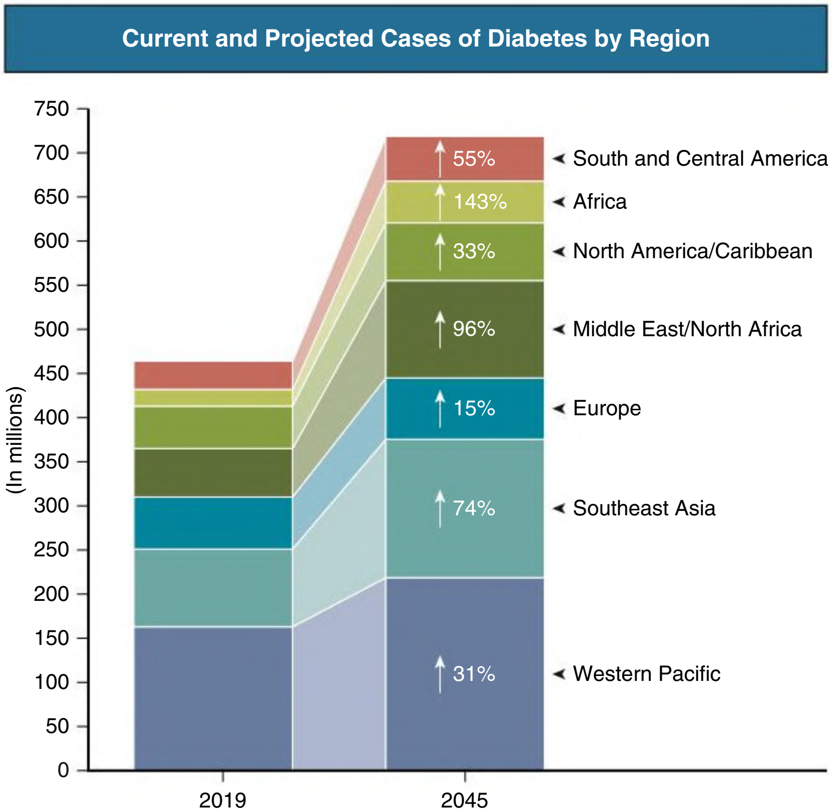

Fig. 31.6 Predicted Increase in Diabetes Prevalence by Geographic Region Over the Next Two Decades. The rate of increase is greatest in the developing world. (Modified from Saeedi P, Petersohn I, Salpea P, et al.; IDF Diabetes Atlas Committee. Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: results from the International Diabetes Federation Diabetes Atlas, 9th edition. *Diabetes Res Clin Pract*. 2019;157:107843.)

of overt diabetes (e.g., hyperglycemia) because of CKD-associated weight loss, impaired kidney gluconeogenesis, or increased insulin half-life in CKD.

The proportion of patients with T1D and T2D who develop proteinuria and elevated serum creatinine concentration is related to the duration of diabetes. Diabetes and its complications caused an estimated 4.2 million deaths worldwide in 2019, of which half occurred in patients below the age of 60, and more than half were caused by increased risks for cardiovascular and other diseases. There is a more rapid increase in the prevalence of T2D in lower-income versus higher-income regions: 4 out of 5 people with diabetes live in low- and middle-income countries, and these countries are predicted to experience the greatest surge in diabetes over the next 25 years (Fig. 31.6).[80] For example, the prevalence of diabetes from 2019 is predicted to increase by 143% in Africa, 74% in Southeast Asia, and 55% in South and Central America by 2045 versus 33% in North America, 15% in Europe, and 31% in the Western Pacific. In Africa and the Middle East/North Africa, 75% and 50% of deaths due to diabetes were in people younger than 60 years, respectively. In Asia, high mortality from diabetes is most prominent in patients aged 50 to 60 years, which translates to a reduction in life expectancy of more than a decade. Up to 60% of Asian patients with diabetes have elevated albuminuria, compared with 30% to 40% reported in Western diabetic populations in cross-sectional surveys.[81]

CLINICAL MANIFESTATIONS AND NATURAL HISTORY

Diabetic kidney disease is part of a generalized microvascular and macrovascular disease.

Obesity, Metabolic Syndrome, and Kidney Disease

The metabolic syndrome—defined by the presence of three or more of certain factors (increased waist circumference, elevated triglycerides, decreased high-density lipoprotein, elevated blood pressure [BP], and elevated fasting blood glucose concentration)—is increasingly recognized as a major contributor to cardiovascular diseases with a negative impact on kidney function.[82,83] Obesity is often defined as body mass index greater than 30, although different values define obesity in other countries (e.g., China, India, Japan). Obese individuals have large kidneys and glomerulomegaly, with increased kidney blood flow, increased filtration fraction, and glomerular hyperfiltration.[84] Obese patients have moderate albuminuria even in the absence of hypertension. The resemblance of obesity-related kidney disease to early DKD is striking. In addition, sleep apnea, which is common in obese individuals, leading to hypoxic episodes, may contribute to kidney impairment. Visceral adipocytes are a potent source of deleterious factors[85] and could have an impact on kidney function (Ang II, leptin, TNF-α). By contrast, the secretion and plasma concentration of adiponectin, an adipokine with cardiovascular protective, antidiabetic, and antiinflammatory properties, are markedly decreased in obesity and its related pathologies (see the section Glomerular Changes). Thus, kidney changes may occur years before the manifestation of T2D during obesity and the development of the metabolic syndrome.

Evolution of Diabetic Kidney Disease

One of the earliest changes of kidney function in patients with T1D and many with T2D is an increase in GFR, or hyperfiltration, which is accompanied by an increase in kidney size. The next observable change is the development from normal (<30 mg albumin/24 h) to *moderately increased albuminuria* (30-300 mg albumin/24 h or 30 to 300 mg/g or 3 to 30 mg/mmol creatinine, graded as A2 by the Kidney Disease: Improving Global Outcomes [KDIGO] nomenclature[86]), previously termed *microalbuminuria*. This may progress to severely increased albuminuria (>300 mg albumin/24 h or >300 mg/g or >30 mg/mmol creatinine; graded as A3 by KDIGO), previously termed *macroalbuminuria* (Table 31.1). Williams[87] proposed a scheme of the natural history and pathophysiology of nephropathy typically observed in T2D over a period of 20 years (Fig. 31.7).

TABLE 31.1 Classification of Albuminuria Categories

	URINE ALBUMIN EXCRETION			
		SPOT URINE		
Condition	24-hr Urine (mg/day)	(mg/g Cr)	(mg/mmol Cr)	Category
Normoalbuminuria	<30	<30	<3	A1
Moderately increased albuminuria	30–300	30–300	3–30	A2
Severely increased albuminuria	>300	>300	>30	A3

Mogensen[88] proposed a scheme of the different stages of DKD that is largely valid in T1D but less reliable in T2D. In the latter, CKD may occur in the absence of albuminuria, possibly as a result of macrovascular disease. Of note, with improved glycemic and BP control there is a growing population of patients with T1D with normoalbuminuria who have progressive DKD.[87] In this cohort, progressive kidney function decline, not albuminuria, is the predominant clinical feature. The putative mechanisms that initiate and sustain progressive kidney function decline in T1D are not well known. In clinical practice, annual eGFR decline is expected to be within 5 mL/min/1.73 m^2 while on a RAS blocker, and further investigation and more frequent monitoring are warranted for attrition rates above this level.

Hypertension and Diabetic Kidney Disease

In patients with T2D, hypertension often precedes the onset of diabetes by years. At diagnosis of T2D, an abnormal BP and an abnormal circadian BP profile are found in 80% of patients. Prediabetic hypertension increases the risk for onset and progression of DKD. The onset of DKD in T2D further increases BP, though this relationship is generally much less than in T1D. The pathogenesis of hypertension in T2D is complex and involves RAS activation, direct sympathetic nerve activation, and macrovascular changes.[89] Furthermore, genetic factors defining primary hypertension as well as diabetes are clustered (Fig. 31.8).

In DKD, nocturnal BP decrease is frequently attenuated, and this nondipping phenomenon may precede the onset of moderately increased albuminuria.[89] Furthermore, BP responses to exercise tend to be exaggerated. Stiffening of the aorta increases the peak systolic pressure and decreases the diastolic pressure, resulting in increased BP amplitude, which explains why isolated systolic hypertension is so common in patients with T2D.[90] Low diastolic pressure increases

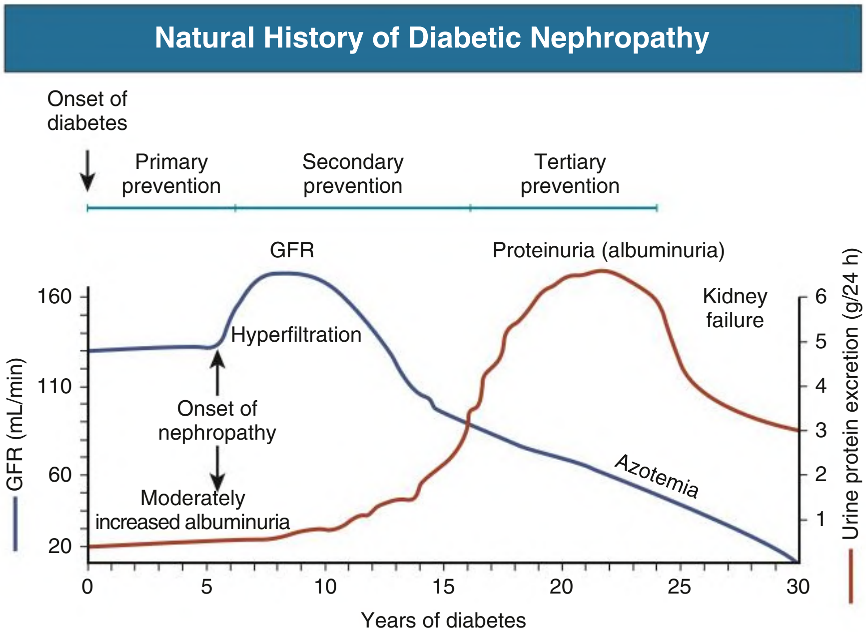

Fig. 31.7 **Natural History of Diabetic Nephropathy.** Changes in glomerular filtration and proteinuria over time from the onset of diabetes. Proteinuria reduction is shown as a tertiary prevention. *GFR,* Glomerular filtration rate. (From Krolewski AS. Progressive renal decline: the new paradigm of diabetic nephropathy in type 1 diabetes. *Diabetes Care.* 2015;38[6]:954–962.)

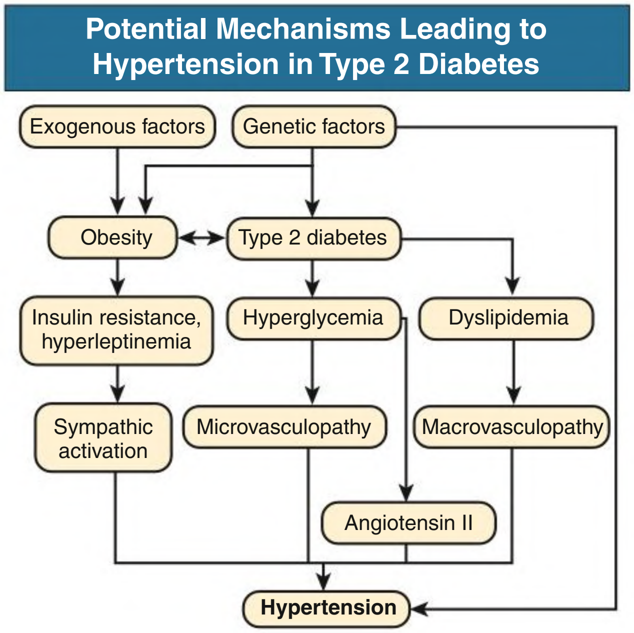

Fig. 31.8 **Overview of Potential Mechanisms Leading to Hypertension in Patients With Type 2 Diabetes.** Genetic susceptibility factors for primary hypertension and diabetes may be clustered so that an individual patient may have a higher incidence of both diseases. Obesity and metabolic syndrome lead to insulin resistance and hyperleptinemia associated with sympathetic nerve activation. Hyperglycemia directly activates the renin-angiotensin system and, in addition, stimulates development of hypertension through kidney microvasculopathy. Dyslipidemia leads to stiffness of vessels and hypertension through macrovasculopathic alteration.

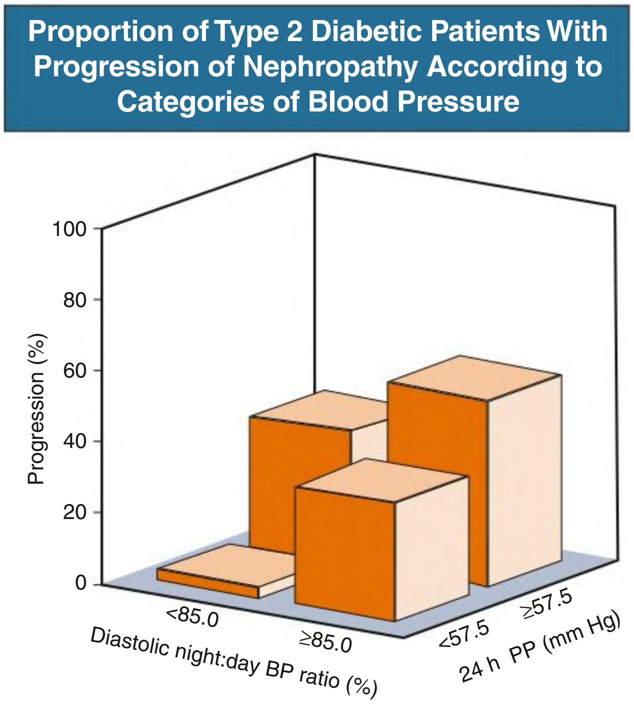

Fig. 31.9 **Proportion of Patients With Type 2 Diabetes With Progression of Nephropathy According to Categories of Blood Pressure (BP).** Progression risk according to categories of nighttime to daytime diastolic BP (median value <85% or ≥85%) and 24-hour ambulatory pulse pressure (PP) (median value <57.5% or ≥57.5%). (From Knudsen ST, Laugesen E, Hansen KW, et al. Ambulatory pulse pressure, decreased nocturnal blood pressure reduction and progression of nephropathy in type 2 diabetic patients. *Diabetologia.* 2009;52:698–704.)

the risk for coronary events because coronary perfusion occurs during diastole only.[91] Augmented pulse pressure and impaired nocturnal BP decline are independent predictors of nephropathy progression in T2D (Fig. 31.9). The increased prevalence of sleep apnea in DKD also contributes to hypertension.[92]

Associated Extrarenal Microvascular and Macrovascular Complications

Diabetic retinopathy is present in virtually all patients with T1D and albuminuria from DN. In contrast, only 50% to 60% of proteinuric patients with T2D have retinopathy.[93,94] Consequently, the absence of retinopathy does not exclude the diagnosis of DN in patients with T2D.[94] In patients with DN, retinopathy tends to progress more rapidly, hence more frequent ophthalmologic monitoring is needed.

Many patients with DN also have polyneuropathy. Sensory polyneuropathy is an important cause of the diabetic foot, which correlates with loss of kidney function. Because cardiac innervation is defective, pain and angina are frequently absent when the patient has coronary heart disease and myocardial infarction. Further consequences of autonomic polyneuropathy are gastroparesis and diarrhea or constipation (see Chapter 89). Erectile impotence and detrusor paresis with delayed and incomplete emptying of the bladder are common urologic problems.

The major macrovascular complications associated with DKD are stroke, coronary heart disease, and peripheral vascular disease,[95,96] with a fivefold higher frequency than diabetic patients without DKD.

Survival in Patients With Diabetic Kidney Disease

The presence of DKD greatly increases mortality in patients with both T1D and T2D. Compared with the general population, mortality in patients with T1D and no proteinuria is elevated by two- to threefold, which increased to 20-fold to 200-fold with proteinuria.[97,98]

The major increase in risks occurs not only with the onset of albuminuria (Fig. 31.10) but also in the upper normal range of albuminuria (Fig. 31.11). Urinary albumin excretion is a good predictor of cardiovascular events.[98] Albuminuria likely reflects generalized endothelial cell dysfunction with an increased risk for atherosclerosis[98] and other associated cardiovascular risk factors, such as elevated BP, dyslipoproteinemia, increased platelet aggregation, and increased C-reactive protein concentration. Other biomarkers for progressive DKD have been described including soluble TNFR1 or TNFR2,[99] and other urine metabolites such as aconitic acid and 3-hydroxyisobutyrate.[45]

KIDNEY PATHOLOGY

After the onset of diabetes, kidney weight increases by an average of 15%. Kidney size remains increased until overt nephropathy is established. Most patients with T1D have a sustained increase in glomerular volume and glomerular capillary luminal volume. These changes are accompanied by hypertrophy of the interstitium.[100]

In patients having diabetes for more than 10 years, regardless of whether nephropathy is present, GBM thickening up to three times the normal range of 270 to 359 nm is common (Fig. 31.12). In advancing DKD, there is a consistent correlation between GBM thickness and fractional mesangial volumes with albuminuria.

Nodular glomerular intercapillary lesions in advanced DKD were described in 1936 by Kimmelstiel and Wilson (Fig. 31.13C).[101] The nodules are located in the central regions of peripheral glomerular lobules as well-demarcated eosinophilic and periodic acid–Schiff–positive masses (see Fig. 31.13C–D). When they are not acellular, nodules contain pyknotic nuclei. It is suggested that nodules result from microaneurysmal dilation of the associated capillary followed by mesangiolysis and laminar organization of the mesangial debris with lysis of the center of the lobule. Foam cells often surround the nodules. These appearances, reported in only 10% to 50% of biopsy specimens in both T1D and T2D, are also seen in diseases with a membranoproliferative glomerulonephritis pattern (see Chapter 22), in amyloidosis and light-chain deposition disease (see Chapter 28), and specific stains and immunofluorescence findings, respectively, will clarify the diagnosis.

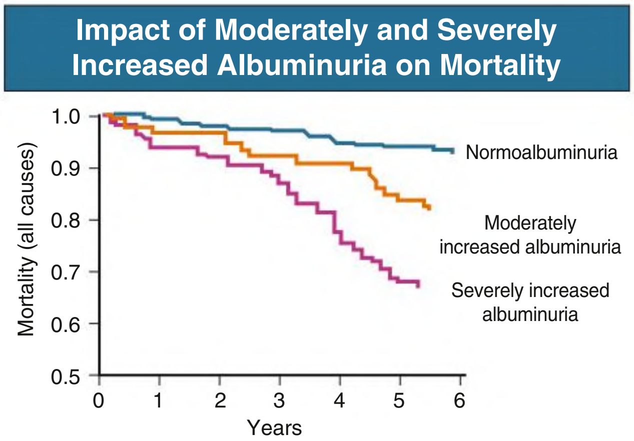

Fig. 31.10 **Influence of Microalbuminuria and Macroalbuminuria on Mortality.** The influence of microalbuminuria and macroalbuminuria on mortality was evaluated prospectively in 328 White patients with non–insulin-dependent diabetes mellitus observed for 5 years. Microalbuminuria and macroalbuminuria led to a significant increase in total mortality compared with that in patients who remained normoalbuminuric. (From Gall MA, Borch-Johnsen K, Hougaard P, et al. Albuminuria and poor glycemic control predict mortality in NIDDM. *Diabetes.* 1995;44:1303–1309.)

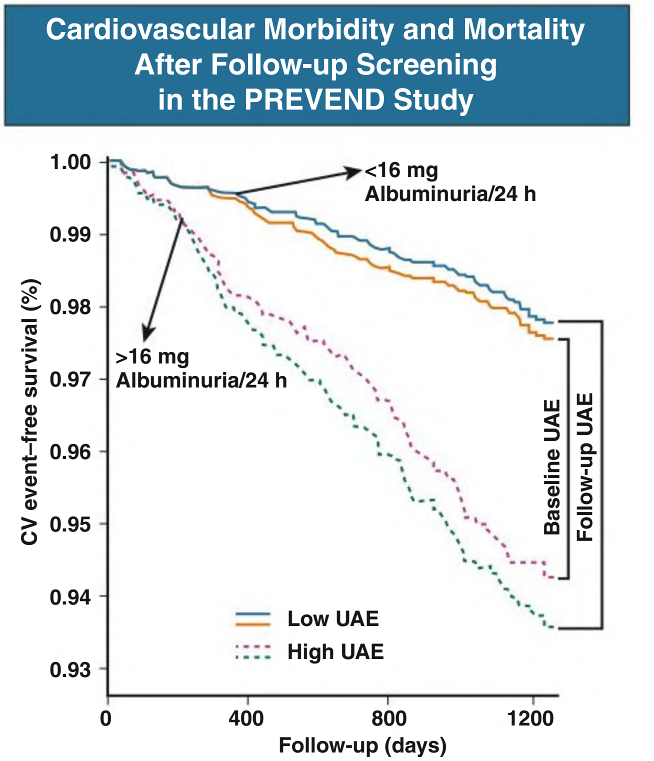

Fig. 31.11 **Event-Free Survival for Cardiovascular (CV) Morbidity and Mortality After Follow-up Screening in the PREVEND Study.** Individuals are stratified according to the presence of a high or low urinary albumin excretion (UAE). High and low UAE are defined by either the UAE measurement from the baseline screening *(orange and purple lines)* of approximately 4.2 years before follow-up screening or the repeated UAE measurement at time of the follow-up screening *(blue and green lines)*. To allow comparison, the survival curves for the 6800 individuals with either the baseline or follow-up measurement of UAE are plotted in the same graph. A high UAE *(dashed lines)* is defined as a UAE ≥16.2 mg/24 h, being the 75th percentile of UAE with use of the UAE measurement of the baseline screening. (From Brantsma AH, Bakker SJ, de Zeeuw D, et al; PREVEND Study Group. Extended prognostic value of urinary albumin excretion for cardiovascular events. *J Am Soc Nephrol.* 2008;19:1785–1791.)

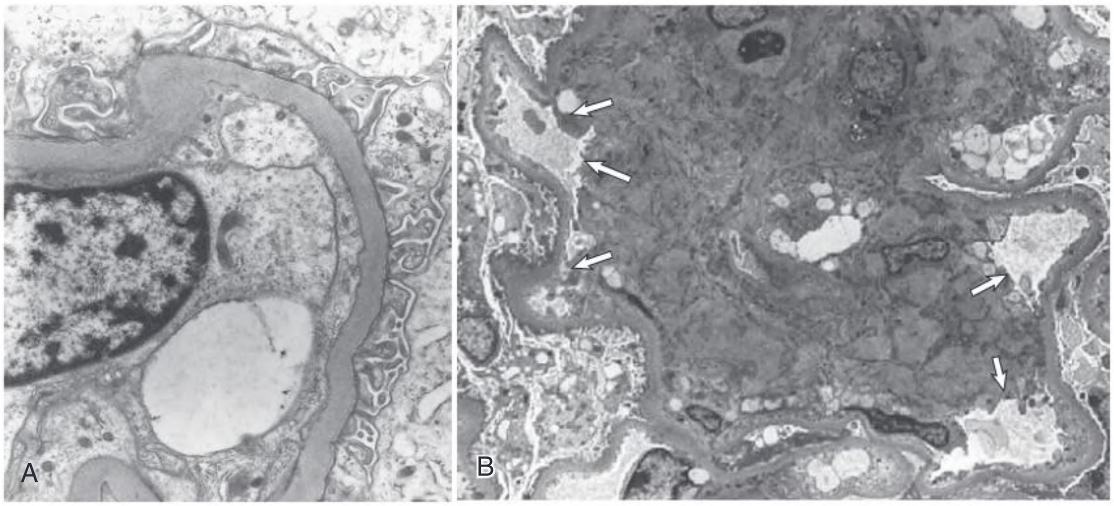

Fig. 31.12 **Electron Microscopy of Structural Changes in Diabetic Nephropathy.** (A) Glomerular basement membranes are diffusely thickened. (B) The expanded mesangium encroaches on the capillary spaces *(arrows)*.

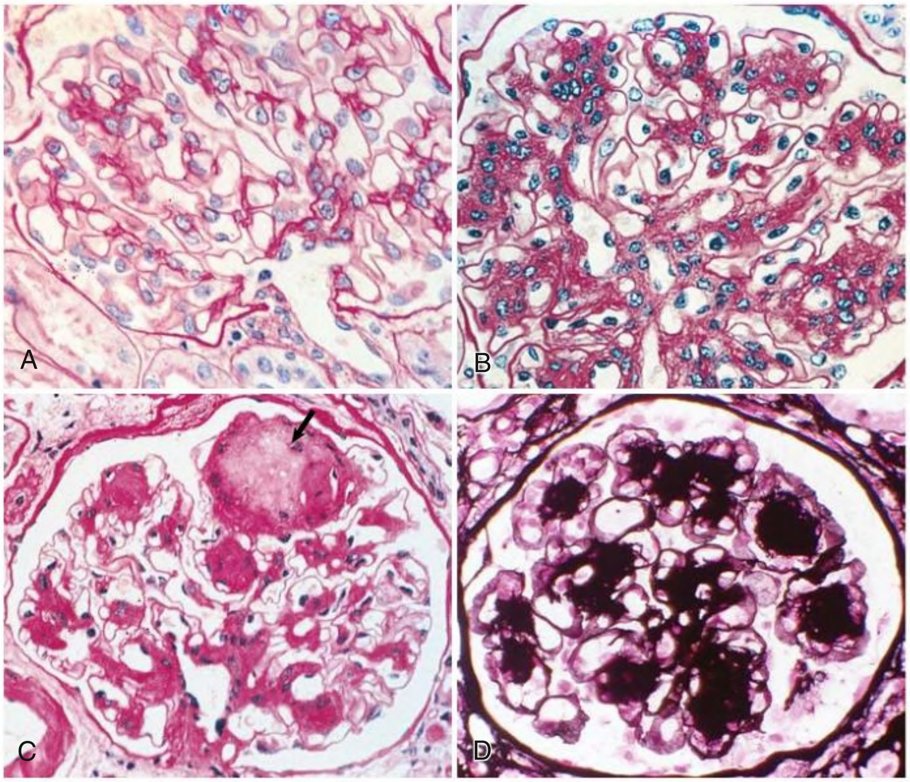

Fig. 31.13 Light Microscopy of Glomerular Changes in Diabetic Kidney Disease. (A) Normal glomerulus. (B) Diffuse glomerular lesion. Widespread mesangial expansion. (C) Nodular lesion and mesangial expansion. There is a typical Kimmelstiel-Wilson nodule at the top of the glomerulus *(arrow).* (D) Nodular lesion. Methenamine silver staining shows the marked nodular expansion of mesangial matrix. (A–C, Periodic acid–Schiff reaction.)

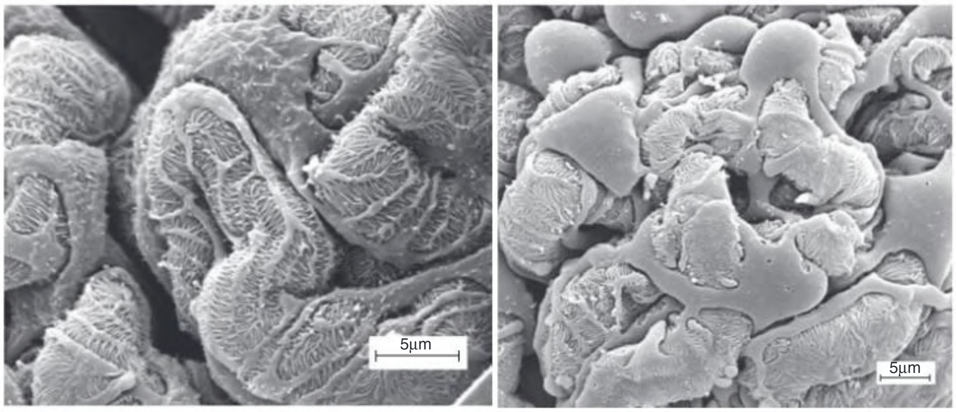

Fig. 31.14 Electron Micrograph of External Surface of Glomerular Tufts From Rats After Removal of Bowman's Capsule by Freeze Fracture. *Left,* Normal rat kidney with podocyte cell body; the primary processes and terminal foot processes resting on the glomerular capillary basement membrane are seen clearly. *Right,* In the diabetic rat kidney, the decrease in the density of foot processes and the denuded glomerular capillary basement membrane are apparent. (From Marshall SM. The podocyte: a major player in the development of diabetic nephropathy? *Horm Metab Res.* 2005;37[suppl 1]:9–16.)

Diffuse glomerular lesions occur more often than nodular lesions, seen in 90% of patients with T1D greater than 10 years and in 25% to 50% of T2D patients. They consist of an increase of mesangial matrix extending to involve the capillary loops (see Fig. 31.13B). In contrast to nodular lesions, which are of little functional significance, the degree of diffuse glomerulosclerosis and in particular mesangial matric expansion correlates with worsening kidney function.[102] In more severe disease, capillary wall thickening and mesangial expansion lead to capillary narrowing (see Fig. 31.12B) and hyalinization, with accompanying periglomerular fibrosis.

Podocytes are involved early in the course of DKD in T1D and T2D (Fig. 31.14), and an increase in foot process width is already observed with only slight increases in albuminuria.[103,104] Longitudinal studies demonstrated a reduction in podocyte number that closely correlated with proteinuria.[103]

Arteriolar lesions are prominent in diabetes. Hyaline material progressively replaces the entire wall structure and involves both the afferent and efferent vessels, which is highly specific for diabetes.

A new classification of DN was introduced in 2007. This classification considers not only glomerular changes (Fig. 31.15) but also pathologic alterations of the tubulointerstitium and vasculature.[105] Immunohistologic examination is usually negative, but linear immunoglobulin G (IgG) can be seen occasionally because of passive trapping in the GBM (Fig. 31.16).

Tubulointerstitial fibrosis and tubular atrophy may be the best pathologic correlates for progressive decline in GFR. Tubulointerstitial fibrosis and renal arteriolosclerosis are more prevalent in T2D than

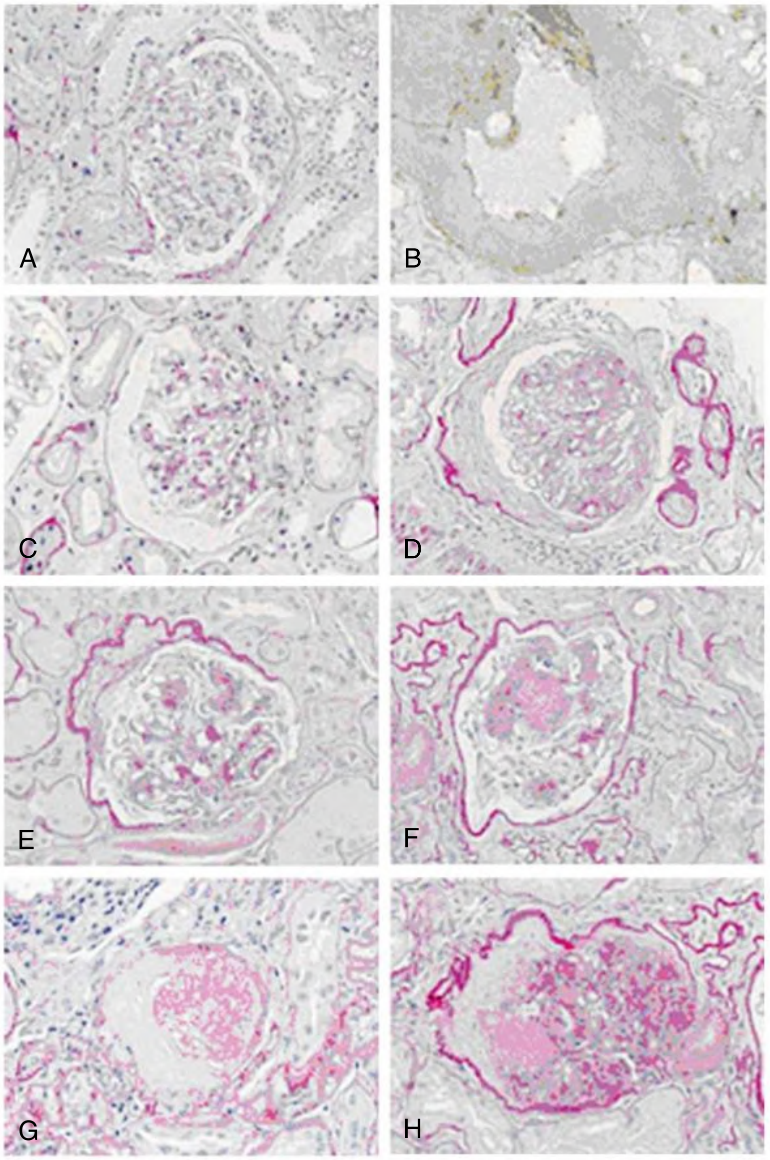

Fig. 31.15 Pathologic Classification of Diabetic Kidney Disease (DKD). Representative examples of the morphologic lesions in DKD. (A) Glomerulus showing only mild ischemic changes, with splitting of Bowman's capsule. No clear mesangial alteration. (B) Electron micrograph of this glomerulus. The mean width of the glomerular basement membrane was 671 nm (mean taken over 55 random measurements). Electron microscopy provides the evidence for classifying the biopsy with only mild light microscopy changes into class I. (C–D) Class II glomeruli with mild and moderate mesangial expansion, respectively. In part C, the mesangial expansion does not exceed the mean area of a capillary lumen (IIa), whereas in part D it does (IIb). (E–F) A class III Kimmelstiel-Wilson lesion is seen in part F. The lesion in part E is not a convincing Kimmelstiel-Wilson lesion; therefore, on the basis of the findings in this glomerulus, the finding is consistent with class IIb. For the purpose of the classification, at least one convincing Kimmelstiel-Wilson lesion (as in F) needs to be present. (G) Example of glomerulosclerosis that does not reveal its cause (glomerulus from same biopsy as H). (H) Signs of class IV DKD consist of hyalinosis of the glomerular vascular pole and a remnant of a Kimmelstiel-Wilson lesion on the opposite site of the pole. For the purpose of the classification, signs of DKD should be histopathologically or clinically present to classify a biopsy with global glomerulosclerosis in more than 50% of glomeruli as class IV. (From Tervaert TW, Mooyaart AL, Amann K, et al. Pathologic classification of diabetic nephropathy. *J Am Soc Nephrol.* 2010;21:556–563.)

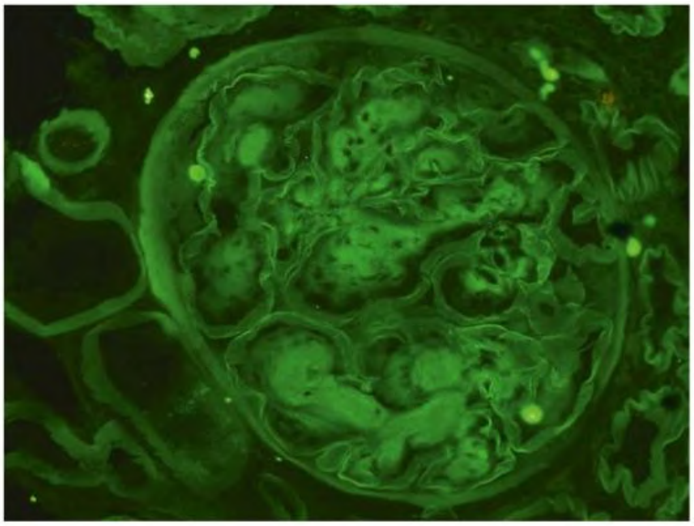

Fig. 31.16 Immunofluorescence for Glomerular Immunoglobulin G (IgG) in Diabetic Nephropathy. Faint staining of the glomerular basement membrane (GBM) for IgG results from passive trapping of IgG in the expanded GBM. (Courtesy Prof. Peter Furness, Leicester, UK.)

T1D. In fact, kidney structure is heterogeneous in patients with T2D; only a subset of patients with T2D have typical diabetic glomerulopathy, whereas others have more advanced tubulointerstitial and vascular rather than glomerular lesions or have normal or near-normal kidney structure,[105] or even features suggestive of glomerular ischemia.

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

The diagnosis of DKD is based on the detection of proteinuria. In addition, most patients also have hypertension and retinopathy. The main evaluation procedures in the patient with suspected DKD include:

- Measurement of urinary albumin or protein
- Measurement of serum creatinine concentration and estimation of GFR
- Measurement of BP
- Ophthalmologic examination

Measurement of Albuminuria or Proteinuria

There is substantial individual day-to-day variation in albumin excretion (coefficient of variation, 30%–50%) and day and night differences (Fig. 31.17). Even in the upper quantiles of so-called normoalbuminuria, the risk for progression and cardiovascular events is elevated. At concentrations of 30 to 300 mg/24 h, albumin is normally not detected by conventional dipsticks for proteinuria. Albumin can be detected by special dipsticks for moderately increased albuminuria or by a suitable laboratory assay. A first-void morning urine sample is preferred. The normal range is less than 20 μg/mL.

The detection of urinary albumin is a specific indicator of DN only if confounding factors such as fever, physical exercise, urinary tract infection, nondiabetic kidney disease, hematuria from other causes, heart failure, uncontrolled hypertension, and uncontrolled hyperglycemia have been excluded.[106]

The main advantage of testing for moderately increased albuminuria early is that it predicts a high kidney and cardiovascular risk and thus allows targeted intervention. The American Diabetes Association, KDIGO, and others recommend annual testing of albumin-to-creatinine ratio (more costly) or protein-to-creatinine ratio.

By definition, there is clinically overt DKD (or severely increased albuminuria) if the rate of albumin excretion exceeds 300 mg/day. At this point, serum proteins other than albumin are usually excreted in the urine as well (nonselective proteinuria).

Measurement of Blood Pressure

The following points should be noted:

- In overweight T2D patients, the size of the cuff should be adapted to the upper arm circumference. When this exceeds 32 cm, cuffs of 18 cm width are indicated.
- Patients with severe autonomic neuropathy tend to develop orthostatic hypotension, defined as a decrease of systolic BP by more than 20 mm Hg in the upright position.
- The circadian BP profile tends to be abnormal in the early stages, and even a paradoxical increase in nocturnal BP is not rare. In DKD patients, increase in nocturnal BP is independently associated with a 20-fold higher mortality and a higher risk for kidney failure. Ambulatory BP recordings are useful to assess the efficacy of antihypertensive treatment.
- In diabetic patients with sclerosis or calcification of the radial and brachial arteries, there may be pseudohypertension characterized by spuriously elevated BP despite normotension upon intraarterial BP measurements with a discrepancy between mild target organ damage (e.g., left ventricular hypertrophy) and very high measured BP values. Such patients tend to develop marked hypotension even with modest antihypertensive therapy.

Measurement of Serum Creatinine and Estimation of Glomerular Filtration Rate

Serum creatinine concentration may be inaccurate in cachexic patients with low muscle mass. This problem is particularly frequent in elderly patients with T2D. KDIGO recommends reporting estimated GFR in adults using the 2009 CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) creatinine equation.

Differential Diagnosis

Although hematuria is one of the atypical features indicating the presence of nondiabetic kidney disease in patients with diabetes, it may be present in DKD. Moreover, a study identified hematuric patients with pathologically defined DKD, who had significantly lower kidney function than nonhematuric patients with DKD.[107] The prevalence of nephrotic

Circadian Variation of Urinary Albumin Excretion

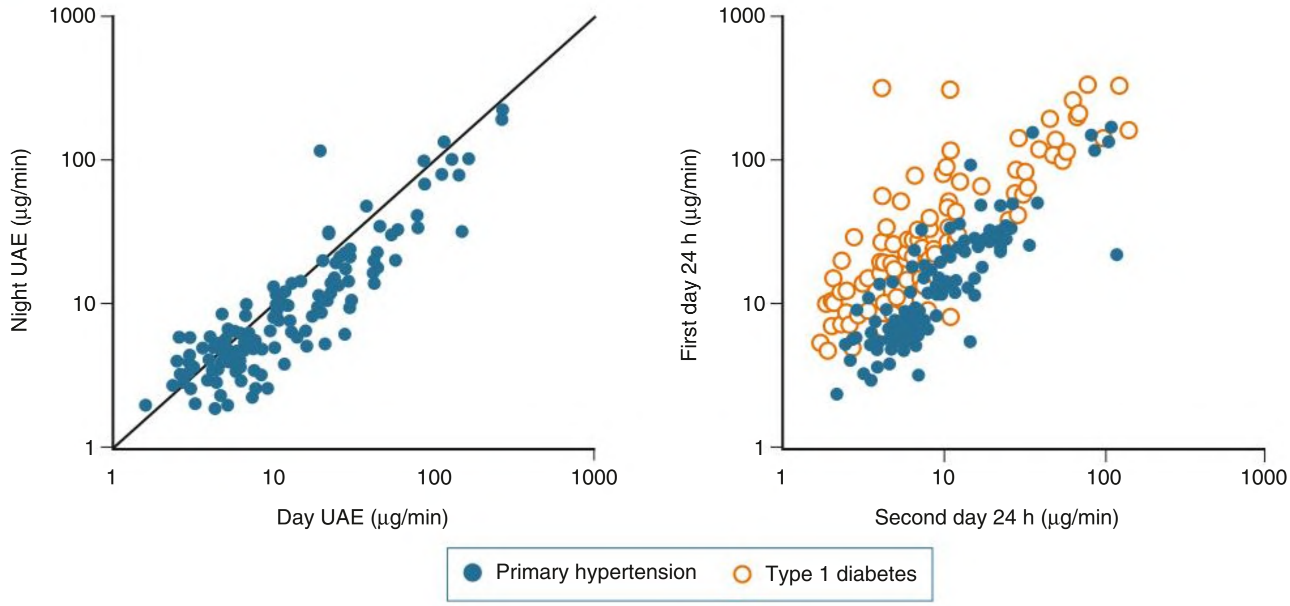

Fig. 31.17 Circadian Variation of Urinary Albumin Exretion (UAE). UAE is lower in resting conditions at night *(left)* than during daytime activity *(right)*. Relationship between UAE assessed on two different days 1 week apart in patients with type 1 diabetes *(open circles)* and primary hypertension *(closed circles)*. There is substantial individual day-to-day variation of albumin excretion and also between day and night collections. (From Redon J. Measurement of microalbuminuria: what the nephrologist should know. *Nephrol Dial Transplant.* 2006;21:573–576.)

Pathologic Diagnoses Other Than Diabetic Nephropathy

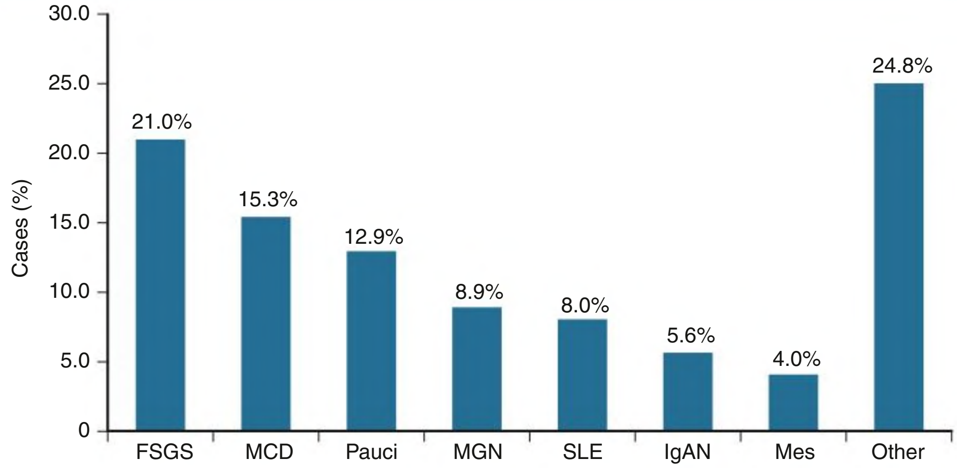

Fig. 31.18 Pathologic Diagnoses Other Than Diabetic Nephropathy Are Found in More Than Half of Patients With Type 2 Diabetes With Proteinuria. A total of 233 patients were studied; 53.2% (124 patients) had a diagnosis of nondiabetic kidney disease. *FSGS,* Focal segmental glomerulosclerosis; *IgAN,* immunoglobulin A nephropathy; *MCD,* minimal change disease; *Mes,* mesangial immune complex glomerulonephritis; *MGN,* membranous nephropathy; *Pauci,* ANCA-positive pauci-immune glomerulonephritis; *SLE,* systemic lupus erythematosus. (From Pham TT, Sim JJ, Kujubu DA, et al. Prevalence of nondiabetic renal disease in diabetic patients. *Am J Nephrol.* 2007;27:322–328.)

syndrome and retinopathy was significantly higher in hematuric patients than in nonhematuric patients with diabetic glomerulosclerosis.

On the other hand, other forms of kidney disease may be found in patients with T2D. Younger patients with diabetes, shorter duration of diabetes, and proteinuria without retinopathy may suggest nondiabetic kidney disease.[108] Membranous nephropathy, FSGS, acute interstitial nephritis, postinfectious GN, and IgA nephropathy have all been described in patients with T2D in whom DKD was clinically suspected (Fig. 31.18).

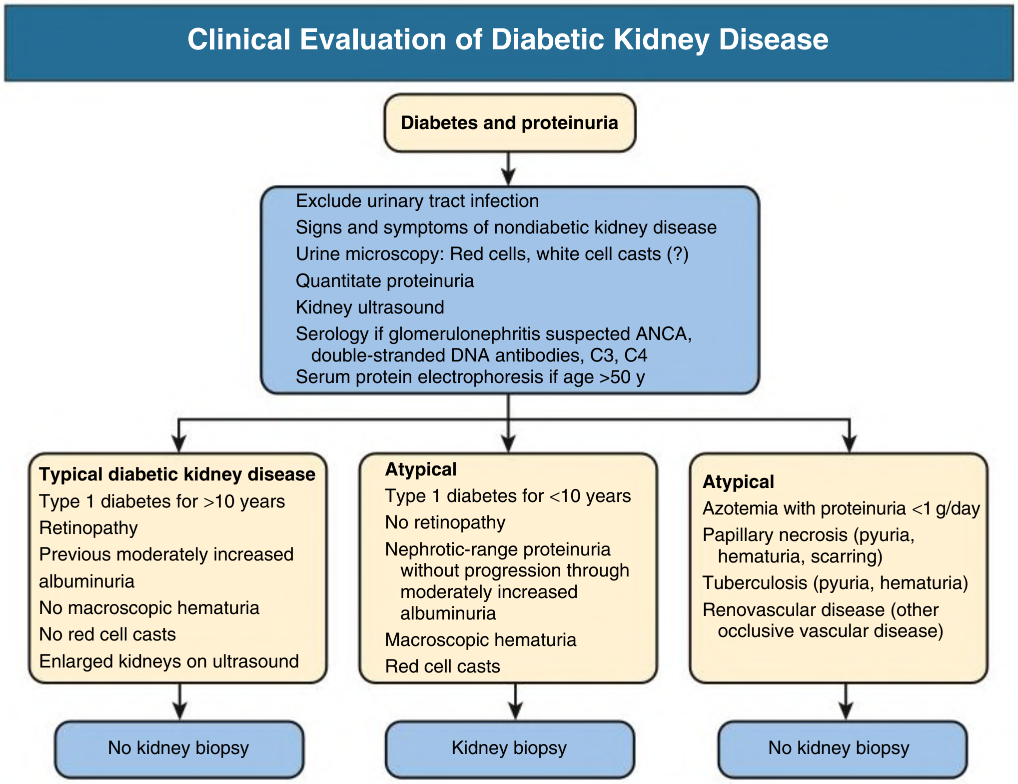

Fig. 31.19 Clinical evaluation of diabetic kidney disease. *ANCA,* Antineutrophil cytoplasmic antibody.

Indications for Kidney Biopsy

Further investigation, including kidney biopsy, should be considered in the following situations[109] (Fig. 31.19):

- If retinopathy is not present in T1D with proteinuria or moderately impaired kidney function (absence of retinopathy does not exclude DKD in T2D).
- If the onset of proteinuria has been sudden and rapid, and the duration of T1D less than 5 years. Alternatively, if the evolution has been atypical, for example, without transition through the usual stages, particularly the development of nephrotic syndrome without previous moderately increased albuminuria.
- If there is macrohematuria or an active nephritic urinary sediment including acanthocytes or red blood cell casts; the sediment in DKD typically is not more than occasional erythrocytes.
- If the decline in kidney function is exceptionally rapid after excluding reversible causes of AKI such as infection, drugs, and obstruction, or if kidney dysfunction is found without significant proteinuria (first, renovascular disease must be excluded) (Fig. 31.20).
- If there are signs and symptoms suggestive of a nondiabetic cause for kidney disease.

If kidney ultrasound reveals small kidneys or a significant size difference, it is prudent not to perform a kidney biopsy. Recent initiatives by the National Institutes of Health (NIH)-funded Kidney Precision Medicine Project are assessing biopsies from patients with DN to determine whether a multiomic approach with single cell transcriptomics, proteomics, and spatial metabolomics could provide new insights in humans.

Approach to the Diabetic Patient With Impaired Kidney Function

When seeing a diabetic patient with CKD, the nephrologist should:

- Assess the cause of CKD (acute vs. chronic kidney impairment; DKD vs. alternative causes of kidney damage).

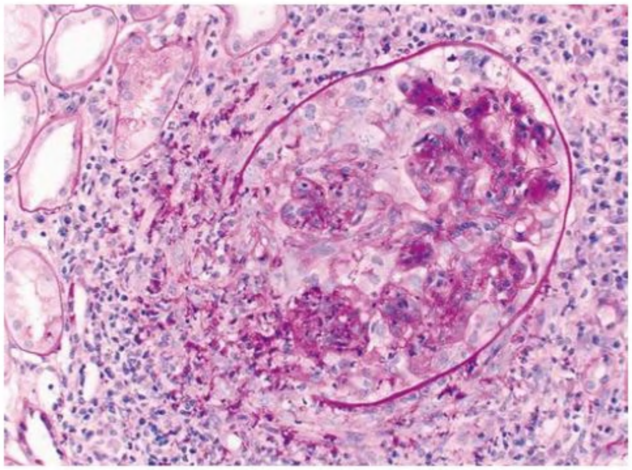

Fig. 31.20 Glomerulonephritis Superimposed on Diabetic Kidney Disease (DKD). A glomerulus showing a cellular crescent with rupture of the Bowman's capsule superimposed on nodular DKD. The patient, known to have diabetic nephropathy, presented with rapidly deteriorating kidney function and red cell casts in the urine.

- Assess the magnitude of proteinuria and the rate of progression.
- Search for typical extrarenal microvascular and macrovascular complications of diabetes.

The majority of diabetic patients with heavy proteinuria or kidney failure have DKD. Kidney ischemia (atherosclerotic kidney artery stenosis or cholesterol embolism) is common in diabetic patients, and many patients with T2D have small kidneys and low GFR without albuminuria, possibly from macrovascular disease.

If urinary tract infection occurs, it is more severe in the diabetic than the nondiabetic patient. Purulent papillary necrosis and intrarenal abscess formation have now become rare.

Diabetic patients with nephropathy are particularly prone to development of AKI after administration of NSAIDs or radiocontrast media or after cardiovascular events or septicemia. Preventive measures for AKI are discussed in Chapter 74. AKI superimposed on preexisting DKD carries a poor kidney prognosis.

SELF-ASSESSMENT QUESTIONS

1. Which of the following statements about diabetic nephropathy is *correct?*
   A. The pathophysiology for diabetic nephropathy is different in T1D and T2D.
   B. Glomerulosclerosis, tubular atrophy, and tubulointerstitial lesions are typical pathologic findings in diabetic nephropathy.
   C. Development of diabetic nephropathy is always associated with microalbuminuria.
   D. Development of diabetic nephropathy is always associated with hypertension.
2. Which of the following is *not* a risk factor for the development of diabetic kidney disease?
   A. Genetic background
   B. Hyperglycemia
   C. Smoking
   D. Anemia
   E. Type of diabetes
3. Which of the following findings *may* suggest a nondiabetic origin of the kidney disease?
   A. Proteinuria greater than 2 g/day
   B. Large kidneys on ultrasound
   C. Increased serum creatinine
   D. Acanthocytes in urine sediment
4. Which of the following observations suggests inflammation is a feature in diabetic kidney disease?
   A. Increased transforming growth factor-β levels
   B. Inflammasome activation in the diabetic kidney
   C. Markedly elevated C-reactive protein and ferritin levels
   D. Peripheral leukocytosis
   E. Presence of urinary red and white cell casts

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